Sébastien Martin, Associate Professor

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Last Updated: 2026-10

SUMMARY

I am an associate professor of Operations at the Kellogg School of Management, Northwestern University. My research focuses on the interface of large-scale optimization and operations management, with applications to transportation, the gig economy, public sector operations and AI.

EDUCATION

Massachusetts Institute of Technology | Cambridge, MA, USA — 2014 — 2019
Ph.D. - Operations Research

Ecole Polytechnique | Palaiseau, France — 2011 — 2015
B.Sc. & M.Sc. - Applied Mathematics

ACADEMIC APPOINTMENTS

Northwestern University - Kellogg | Evanston, IL, USA — 2025-09 — Current
Associate Professor of Operations
I teach the MBA course "AI Foundations for Managers, AgentOps".

Operations Research — 2027-01 — Current
Associate Editor, Transportation Area

Manufacturing & Service Operations Management (MSOM) — 2026-03 — Current
Associate Editor, AI Department

Transportation Science — 2024-10 — Current
Associate Editor

Northwestern University - Kellogg | Evanston, IL, USA — 2020 — 2025-09
Assistant Professor of Operations
I taught the Operations Management core course in the MBA program.

UC Berkeley | Berkeley, CA, USA — 2014-04 — 2014-08
Visiting Researcher

INDUSTRY AND ADVISORY

ESAB Corporation (NYSE: ESAB) | North Bethesda, Maryland, U.S. — 2026-01 — Current
Board Member
Member of the Board of Directors, providing strategic oversight with a focus on innovation, AI adoption, and operational excellence. ESAB is a global leader in fabrication and gas control technology ($3B revenue, 10,000 employees).

ESAB Corporation (NYSE: ESAB) | North Bethesda, Maryland, U.S. — 2025 — 2025-12
AI advisor to the CEO
I served as a strategic advisor to ESAB, working with executive leadership to develop the company's AI strategy, establish its AI steering committee, and train the workforce on leveraging AI agents for innovation and productivity.

Lyft, Inc. | New York City, NY, USA — 2019 — 2020
Postdoctoral Fellow
I worked with the Marketplace Innovation Lab to improve dispatch algorithms.

Google | Mountain View, CA, USA 2016-06 — 2016-08
Software Engineering Intern
Successfully passed the Google Software Engineer coding interviews. Worked for Google Maps. Researched, experimented and implemented novel algorithms to improve maps and navigation data using large geolocation datasets (> 100Gb).

PUBLICATIONS

*Only publicly available papers, in decreasing order of latest update. The current publication status is highlighted in **bold**.*

Published or accepted

[12] Algorithmic Precision and Human Decision: A Study of Interactive Optimization for School Schedules | A. Delarue, Z. Lian, S. Martin
Management Science 2026
Accepted in EC 2024, semi-finalist of the 2024 Wagner Prize.

[11] Trading Flexibility for Adoption: From Dynamic to Static Walking in Ride-Sharing | J. Yan, S. Martin, S. Taylor
Management Science 2025

[10] Detours in Shared Rides | I. Lobel, S. Martin
Management Science 2025

[9] Employees versus Contractors: An Operational Perspective | I. Lobel, S. Martin, H. Song
Manufacturing & Service Operations Management (Frontiers in Operations) 2024

[8] A Better Match for Drivers and Riders: Reinforcement Learning at Lyft | S. Martin and 10+ Lyft collaborators
INFORMS Journal on Applied Analytics 2024
2023 Franz Edelman Award Laureate

[7] Real-Time Rideshare Driver Supply Values using Online Reinforcement Learning | B. Han, H. Lee, S. Martin
KDD 2022 (Machine Learning Conference) 2022

[6] Solving the ride-sharing productivity paradox: Priority dispatch and optimal priority sets | V. Krishnan, R. Iglesias, S. Martin, V. Pattabhiraman, S. Wang, G. van Ryzin
INFORMS Journal on Applied Analytics 2022
Daniel H. Wagner Prize Finalist, 2022

[5] Bus Routing Optimization Helps Boston Public Schools Design Better Policies | D. Bertsimas, A. Delarue, W. Eger, J. Hanlon, S. Martin
INFORMS Journal on Applied Analytics 2020
2019 Franz Edelman Award Laureate

[4] Optimizing schools' start time and bus routes | D. Bertsimas, A. Delarue, S. Martin
Proceedings of the National Academy of Science 2019
Featured in the Wall Street Journal and the Boston Globe. MIT ORC Best Student Paper Award, 2018. Doing Good with Good OR INFORMS award, Second Place, 2019.

[3] Travel Time Estimation in the Age of Big Data | D. Bertsimas, A. Delarue, P. Jaillet, S. Martin
Operations Research 2019

[2] Online Vehicle Routing: The Edge of Optimization in Large-Scale Applications | D. Bertsimas, P. Jaillet, S. Martin
Operations Research 2019
Best Presentation (2018 LIDS conference)

[1] Creating complex congestion patterns via multi-objective optimal freeway traffic control with application to cyber-security | J. Reilly, S. Martin, M. Payer, A. Bayen
Transportation Research Part B: Methodological 2016

In the peer-review process

[9] Navigating the Educational Video Game Paradox | A. Jensen, S. Martin, K. Smilowitz
Major revision, Management Science 2026

[8] Let's Have a Conversation: Designing and Evaluating LLM Agents for Interactive Optimization | J. Drossman, A. Jacquillat, S. Martin
Major revision, Manufacturing & Service Operations Management 2026

[7] The Value of a Little Flexibility in Stable Matching | D. Freund, S. Martin, J. (K.) Zhao
First submission under review, Operations Research 2026
Accepted in EC 2026.

[6] AI homework increases student satisfaction: A field experiment at the Kellogg School of Management | R. Bray, S. Martin
Resubmission under review (major revision), INFORMS Transactions on Education 2026

[5] Two-Sided Flexibility in Platforms | D. Freund, S. Martin, J. (K.) Zhao
Resubmission under review (major revision), Operations Research 2026
MIT ORC Best Student Paper Award, 2024

[4] Labor Cost Free-Riding in the Gig Economy | Z. Lian, S. Martin, G. van Ryzin
Minor revision, Management Science 2026
INFORMS RMP (Revenue Management and Pricing) Student Paper Award Finalist, 2021

[3] Relative Monte Carlo for Reinforcement Learning | A. Bazerghi, S. Martin, G. van Ryzin
Reject and resubmit, Management Science 2026

[2] Human-AI Interactions and Societal Pitfalls | F. Castro, J. Gao, S. Martin
Major revision, Manufacturing & Service Operations Management 2025
Accepted in EC 2024. Featured in the Wall Street Journal & MIT Technology Review.

[1] Value of Sharing in Routing-Scheduling Operations | A. Jacquillat, S. Martin, K. Zhang
Major revision, Management Science 2025

Working papers

[5] The Trap of Complexity in Experimentation | Y. Huang, S. Martin, Z. Qin
Working Paper 2025

[4] Dual-sourcing of capacity | S. Chopra, S. Martin, P. Mishra, K. Smilowitz
Working Paper 2025

[3] Autonomous Vehicles in Ride-Hailing and the Threat of Spatial Inequalities | F. Castro, J. Gao, S. Martin
Working Paper 2024

[2] Supply Prioritization in Hybrid Marketplaces | F. Castro, J. Gao, S. Martin
Working Paper 2023

[1] The Price of Interpretability | D. Bertsimas, A. Delarue, P. Jaillet, S. Martin
Working Paper 2020

RECOGNITIONS

David Besanko Outstanding Professor of the Year Award | Kellogg School of Management, Northwestern University 2026

Kellogg's flagship MBA teaching award, voted annually by the graduating Full-Time and Evening & Weekend MBA classes. Recipients deliver the faculty address at Convocation (linked video).

INFORMS Journal on Computing Meritorious Reviewer Award | INFORMS 2026
For my service as a reviewer for INFORMS Journal on Computing in 2025.

Poets and Quants Best 40 Under 40 MBA Professors | Poets and Quants 2025
Awarded to 40 best business school professors under 40 years old every year

INFORMS Journal on Computing Meritorious Reviewer Award | INFORMS 2025
For my service as a reviewer for INFORMS Journal on Computing in 2024.

Chair Core Course Teaching Award | Kellogg School of Management, Northwestern University
For "teaching excellence" in Kellogg's core course Operations Management 2024

Transportation Science Meritorious Service Award | INFORMS TSL Society 2024
For my service as a reviewer for the journal Transportation Science.

Distinguished Service Award, Operations Research | INFORMS, Operations Research Journal
For "extraordinary reviewer work for the journal". 2024

Best Student Paper Award | MIT ORC 2024
For my paper "Two-Sided Flexibility in Platforms", the student is my co-author Kamessi Zhao.

Franz Edelman Award Laureate (with Lyft) | INFORMS 2023
Most important award for applied operations research, for my work on reinforcement learning with Lyft.

Daniel H. Wagner Prize Finalist | INFORMS 2022
Award for "strong mathematics applied to practical problems", for my work on platform equilibrium optimization with Lyft.

RMP Student Paper Award Finalist | INFORMS 2021
Award for the best student paper in revenue management and pricing for my paper on labor cost free-riding in the gig economy. The student was Zhen Lian.

George B. Dantzig Dissertation Award | INFORMS 2019
For my PhD dissertation. The George B. Dantzig Award is given for the best dissertation in any area of operations research and the management sciences that is innovative and relevant to practice.

TSL dissertation prize | INFORMS Transportation Science and Logistics Society 2019
For my PhD dissertation. Oldest dissertation INFORMS prize, in the general area of transportation science and logistics.

Franz Edelman Award Laureate (with Boston Public Schools) | INFORMS 2019
Most important award for applied operations research, for my work on bus routing optimization with Boston Public Schools.

Doing Good with Good OR award, Second Place | INFORMS 2019
For my paper "optimizing schools' start time and bus routes".

Best Student Paper Award | MIT ORC 2018
For my paper "optimizing schools' start time and bus routes".

Best Presentation | LIDS 2018
For my paper "Online Vehicle Routing: The Edge of Optimization in Large-Scale Applications".

Boston Public Schools Transportation Challenge Winner | Boston Public Schools 2017
Winner of a $30,000 contest to optimize school bus routes and school schedules.

Zodiac Aerospace – Gerondeau Innovation Prize | Zodiac Aerospace & Ecole Polytechnique
Won a €10,000 prize for most innovative start-up, using machine learning to build a smart bicycle that automatically shifts gears. 2013

French Medal of National Defense, Bronze level | France 2012
I received this French military honor for my cumulated time in external operations during my year of service as a military firefighter.

PRESS

Podcast: AI & Fintech — Fintech Frontlines by FinTech @ Kellogg | Spotify 2026
Interview on AI trends in fintech, covering data privacy, credit scoring, and agentic commerce.

Teaching with AI — Talk at Kellogg All Hands Meeting | YouTube 2026
Talk on teaching with AI at the Kellogg School of Management, January 27, 2026.

Building AI foundations for the future | Kellogg News 2026
About my AIML 901 course.

Say Hello to Your New AI Study Buddy | Kellogg Insight 2026
Co-authored with Robert Bray, about our AI homework research.

ESAB Appoints Dr. Sébastien Martin to Board; Announces Retirement of Director Patrick Allender | ESAB Corporation 2025
Press release announcing my appointment to ESAB's Board and Audit Committee.

AI Is Teaching the Next Generation of M.B.A.s the Classic Case Study | Wall Street Journal
Coverage of the first AI case I created for Kellogg. 2025

Sure, AI Can Automate. But How Can You Use It to Innovate? | Kellogg Insight 2025
On using domain expertise and experimentation to build AI case studies.

No Code, No Problem: Kellogg Says Its New AI Course Is Built For Every MBA | Poets & Quants 2025
About the launch of AIML 901 across five departments.

2025 Best 40 Under 40 MBA Professors: Sébastien Martin | Poets & Quants 2025
Recognized among Poets & Quants' 40 Under 40 MBA professors.

The Hidden Costs of Successful Experiments | Kellogg Insight 2025
Explores pitfalls when scaling positive experiment results.

Does Generative AI Impose a Creativity Tax? | MIT Sloan Management Review 2024
Article on creativity trade-offs in generative AI collaboration.

Podcast: When AI Becomes a TA | Kellogg Insight 2024
Podcast conversation about AI teaching assistants.

If AI Gives You Bad Answers, Tell It to Do Better | Wall Street Journal 2024
Discussed research on improving chatbot responses via feedback.

Human-AI Interactions and Societal Pitfalls | Montreal AI Ethics Institute 2024
Essay on the societal risks of human-AI collaboration.

The Big Trade-off at the Heart of Generative AI | Kellogg Insight 2023
Reprinted in Fortune China, Exame, América Economia, and World Economic Forum.

Podcast: Using AI Comes with a Trade-off. Now Multiply That by 8 Billion. | Kellogg Insight 2023
Podcast episode on systemic consequences of widespread AI adoption.

AI from AI: a Future of Generic and Biased Online Content? | UCLA Anderson Review 2023
Interview on risks of AI-generated online content.

At Their Best, Self-Learning Algorithms Can Be a "Win-Win-Win" | Kellogg Insight 2023
Highlights benefits of self-learning algorithms when aligned with stakeholders.

Podcast: What One School District's Fiasco Says About the Strengths and Limits of AI | Kellogg Insight 2022
Podcast episode on lessons from school district AI deployment.

Building an O.R. framework to drive school district policies | ORMS Today 2022
Feature on operations research for equitable school schedules.

I helped design the 2021-22 school schedule changes in San Francisco | SFUSD 2021
Includes project site schoolbelltimes.org.

2 Ways to Improve Rideshares for Drivers and Customers | Kellogg Insight 2021
Strategies to reduce rideshare costs, emissions, and pickup times.

Researchers Designed an Algorithm to Save Schools Money and Improve Equity. The District Loved it. Then Things Got Messy. | Kellogg Insight 2020
Case study on implementation challenges in school transportation optimization.

The Equity Machine | The Boston Globe 2018
Interactive feature on Boston Public Schools bus routing.

What the Boston School Bus Schedule Can Teach Us About AI | Wired 2018
Story on AI-informed scheduling in Boston.

A School Bus Trip to the Crossroads of Policy and Optimization | SIAM News 2017
Profile of optimization work for school buses.

2 MIT Engineers Use Math To Plot A Path For Boston's School Buses | WBUR 2017
Radio coverage on redesigning Boston bus routes.

How Do You Fix a School-Bus Problem? Call MIT | Wall Street Journal 2017
Feature on the Boston bus scheduling project.

LANGUAGES English *(Fluent)* , French *(Native speaker)* , Spanish *(Intermediate)*